

I. Introduction

- Secret message transmission between 2 nodes in presence of eavesdropper is long-standing problem in wireless network security.
- OFDM (Multi-Carrier) is widely used protocol for modern wireless systems, especially in urban and high-scattering regions.
- Possibility of achieving +ve *secrecy rate* in static environments against strong eavesdroppers (more antennas and better SNR), with or without channel reciprocity is recently revealed in [1-3].
- STEPP[4] is a round-trip communication scheme that:
 - Guarantees +ve secrecy rate (unless Eve's channel is not noiseless) even if: a) Eve has more antennas than both users, b) Eve has better SNR, c) Channel between user is not reciprocal
 - Is asymptotically (∞ Tx power) optimal [4]

II. Main Contribution

- Discuss application of STEEP in Multi-Carrier (OFDM) setup (referred to as MC-STEPP) to improve 'Average Achievable Secrecy Rate' (AASR)
- Propose carrier pairing policy which is shown to be optimal
- Power scheduling at nodes
- Simulation results to show significant improvement w.r.t. AASR compared to classic W.T.C. scheme

III. System Model

- Alice & Bob are Half-Duplex single antenna nodes
 - Eve has $N_E \geq 1$ antennas
 - # of carriers: N_C
 - Flat fading non-reciprocal channel; assume all CSI to be i.i.d. standard complex Gaussian
 - Large-scale fading at Eve from Alice and Bob: α, β (large-scale fading between Alice and Bob is normalized to unity)
 - All receive channels known to receiver through channel sounding
 - All noise power normalized to unity
 - Transmit power at Alice and Bob in n^{th} carrier: $p_{A,n}, p_{B,n}$
- We can write the SNRs as:

$$\text{SNR at Alice: } b_n = p_{B,n} |h_{AB,n}|^2 \quad (1a)$$

$$\text{SNR at Bob: } a_n = p_{A,n} |h_{BA,n}|^2 \quad (1b)$$

$$\text{SNR at Eve from Alice: } c_n = \alpha p_{A,n} \|\mathbf{g}_{A,n}\|^2 \quad (1c)$$

$$\text{SNR at Eve from Bob: } d_n = \beta p_{B,n} \|\mathbf{g}_{B,n}\|^2 \quad (1d)$$

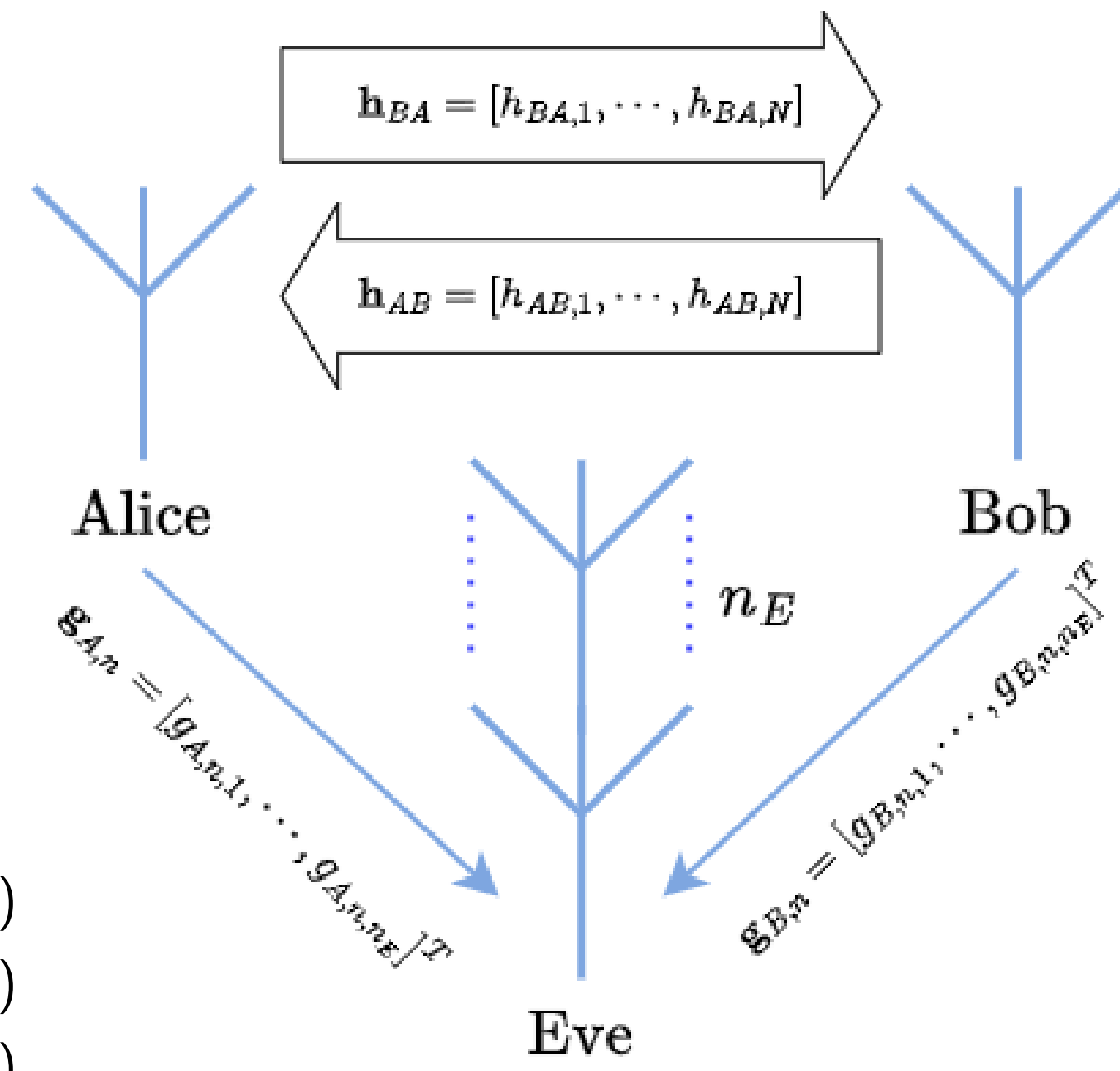


Figure: System Model

IV. Brief Description of STEEP

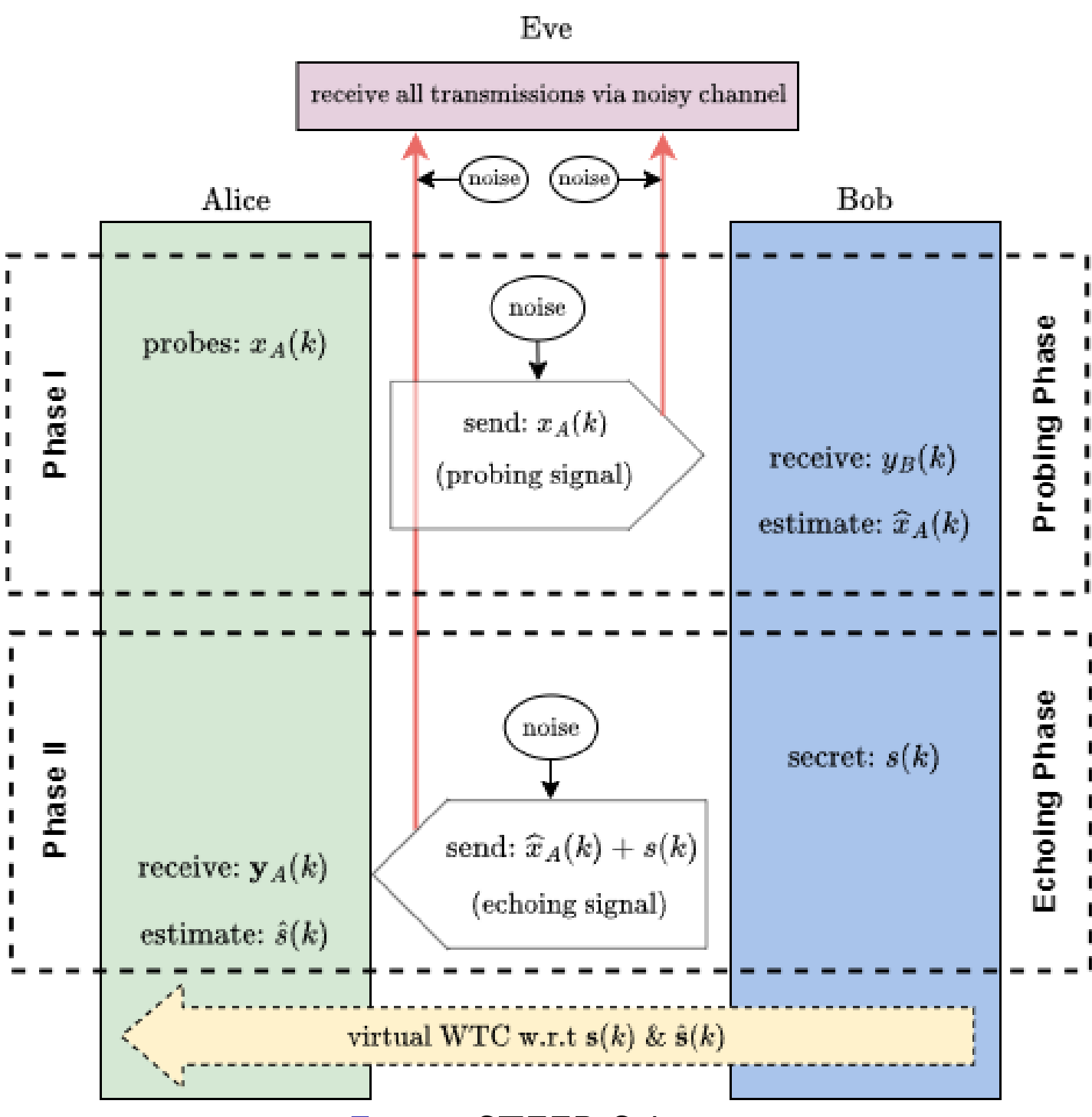


Figure: STEPP Scheme

All $x_{A,n}(k), s_n(k)$ are i.i.d. standard complex Gaussian

STEPP; 2-way roundtrip scheme consisting of two phases: **1) Phase-I** or Probing phase; **2) Phase-II** or echoing phase. Let, secret being sent from Bob to Alice.

Probing Phase:

- Alice sends rand. symbols (probes) $x_{A,n}(k)$ to Bob
- Bob obtains estimate $\hat{x}_{A,n}(k)$ from the received signal

Echoing Phase:

- Bob constructs echo signal $\hat{x}_{A,n}(k) + s_n(k)$ using the secret message $s_n(k)$
- Bob sends this echo signal to Alice
- Alice estimates $\hat{s}_n(k)$

A Virtual W.T.C. formed w.r.t. $s_n(k)$ from Bob to Alice

phase-1:

$$\text{Bob receives: } y_{B,n} = h_{BA,n} x_{A,n} + w_{B,n} \quad (2a)$$

$$\text{Eve receives: } y_{EA,n} = \mathbf{g}_{A,n} x_{A,n} + \mathbf{w}_{EA,n} \quad (2b)$$

phase-2:

$$\text{Alice receives: } y_{A,n} = h_{AB,n} (\hat{x}_{A,n} + s_n) + w_{A,n} \quad (2c)$$

$$\text{Eve receives: } y_{EB,n} = \mathbf{g}_{B,n} (\hat{x}_{A,n} + s_n) + \mathbf{w}_{EB,n} \quad (2d)$$

We define Average Achievable Secrecy Rate (AASR) for MC-STEPP and classic W.T.C.

$$C_{s,steep} \doteq \frac{1}{N} \sum_{n=1}^N \left[\log \left(1 + \frac{b_n}{\frac{a_n b_n}{(a_n+1)^2} + 2} \right) - \log \left(1 + \frac{d_n}{\frac{d_n a_n (a_n + c_n + 1)}{(a_n+1)^2 (c_n+1)} + 2} \right) \right]^+ \quad (3)$$

$$C_{s,classic} = \frac{1}{N} \sum_{n=1}^N \{ [\log(1 + a_n) - \log(1 + c_n)]^+ + [\log(1 + b_n) - \log(1 + d_n)]^+ \} \quad (4)$$

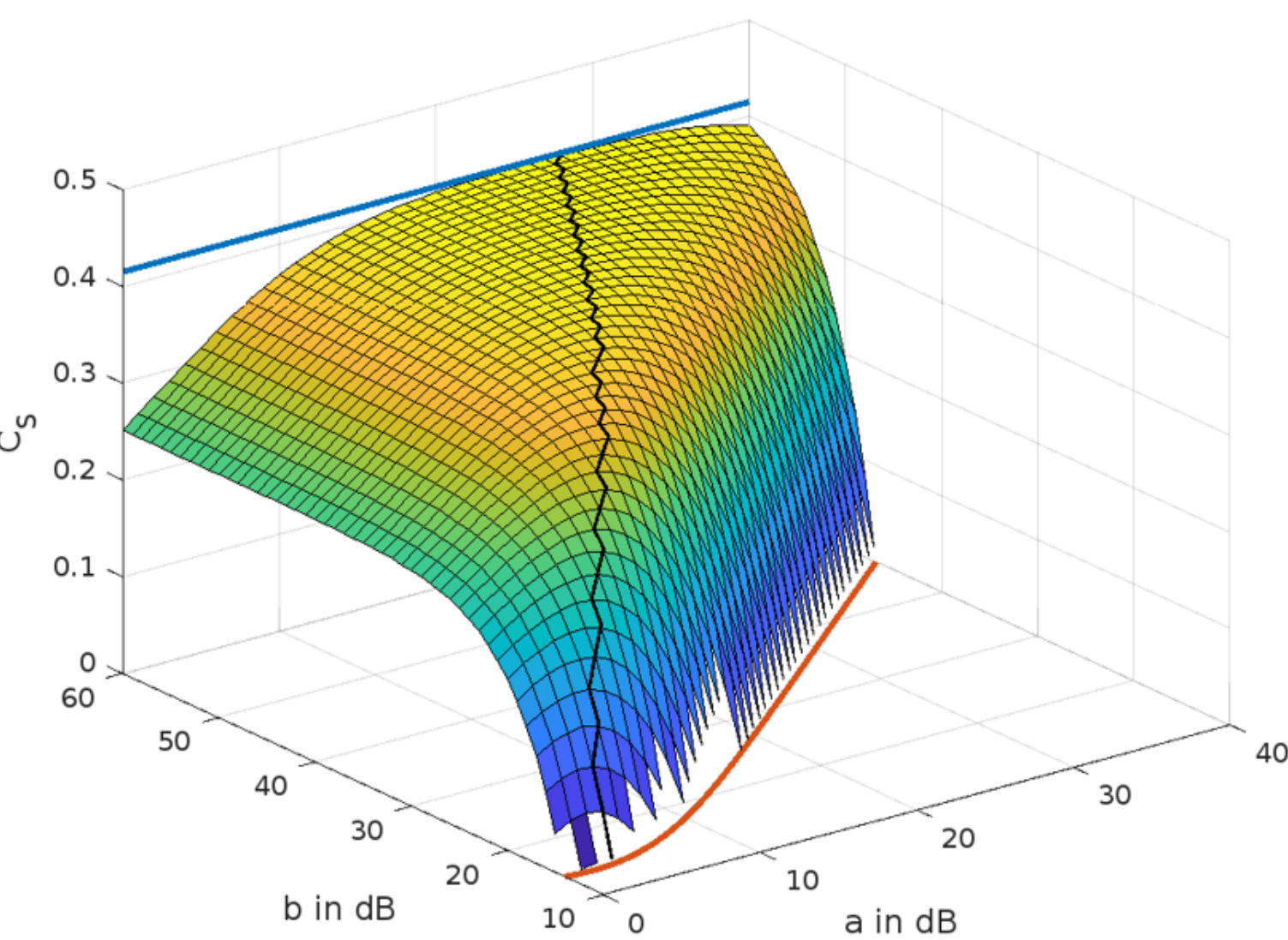


Figure: Secrecy of STEEP

- $c = \alpha a; d = \beta b$
- +ve secrecy if b is greater than threshold (red line)
- Secrecy is monotonically decreasing w.r.t. b (echoing power)
- For a given b (echoing power) there is an optimal a (probing power); (black line)

In this figure $\alpha = 3$ and $\beta = 5$ are used.
In this case classic WTC would yield zero secrecy

V. Carrier Pairing to Improve Secrecy Rate for MC-STEPP; policy-1

- The N_C probing carriers and N_C echoing carriers can be paired in different way.
- Let, for a given roundtrip transmission probing- n is done in carrier i_n and echoing- n is done in carrier j_n .
- Optimal Solution: pair strongest probing carrier with strongest echo carrier and so on.
- $|h_{BA,i_1}| \geq |h_{BA,i_2}| \geq \dots \geq |h_{BA,i_N}|$ and $|h_{AB,j_1}| \geq |h_{AB,j_2}| \geq \dots \geq |h_{AB,j_N}|$.

$ h_{BA,n} $	$ h_{AB,n} $	$ h_{BA,n} $	$ h_{AB,n} $
2.098609	0.716093	2.444616	2.207323
1.350034	0.696184	2.098609	1.60095
1.194074	1.60095	2.01795	1.588619
1.212288	1.049719	1.661328	1.210551
0.445492	1.210551	1.657375	1.049719
0.347562	1.588619	1.350034	0.716093
1.657375	2.207323	1.212288	0.696184
1.661328	0.412482	1.194074	0.470937
2.01795	0.470937	0.445492	0.412482
2.444616	0.098696	0.347562	0.098696

Figure: Carrier Pairing

VI. Power Scheduling at Bob; policy-2

- Total power budget $N_C p_B$ for N_C carriers, power at carrier- n is $p_{B,n}$
- For a given carrier n :

$$v_n(p_{B,n}) \doteq \log \left(1 + \frac{q_n p_{B,n}}{r_n p_{B,n} + 2} \right) - l_n \quad (5)$$

$$g_n(p_{B,n}) \doteq \frac{\partial v_n(p_{B,n})}{\partial p_{B,n}} = \frac{\log e}{1 + \frac{q_n p_{B,n}}{r_n p_{B,n} + 2}} \frac{q_n}{(r_n p_{B,n} + 2)^2} \quad (6)$$

- $q_n = |h_{AB,n}|^2, a_n = p_{A,n} |h_{BA,n}|^2, c_n = \alpha p_{A,n} \|\mathbf{g}_{A,n}\|^2, r_n = \frac{a_n q_n}{(a_n+1)^2}$; and $l_n = \log \left(1 + \frac{\beta \|\mathbf{g}_{B,n}\|^2}{\frac{\beta \|\mathbf{g}_{B,n}\|^2 a_n (a_n + c_n + 1)}{(a_n+1)^2 (c_n+1)} + \frac{2}{p_{B,n}}} \right)$
- For given set of users CSI ($h_{BA,n}, h_{AB,n}$), transmit powers ($p_{A,n}, p_{B,n}$) and α, β ; an estimate of l_n can be made statistically if the distribution of the eavesdropper CSI are known: $l_n = \mathbb{E}_{\mathbf{g}_A, \mathbf{g}_B} l_n$
- $v_n(p_{B,n})$ is monotonically increasing w.r.t. $p_{B,n}$. Considering q_n, r_n, l_n as constants: formulate the following convex optimization problem which should have a unique solution from its KKT conditions:

$$\mathbb{P}1: \max_{p_{B,n}, \forall n} C_{s,steep}; \text{ s.t. } \sum_{n=1}^N p_{B,n} \leq N p_B \quad (7)$$

We let $J = -C_{s,steep} + \mu (\sum_{n=1}^N p_{B,n} - N p_B)$. The KKT conditions are:

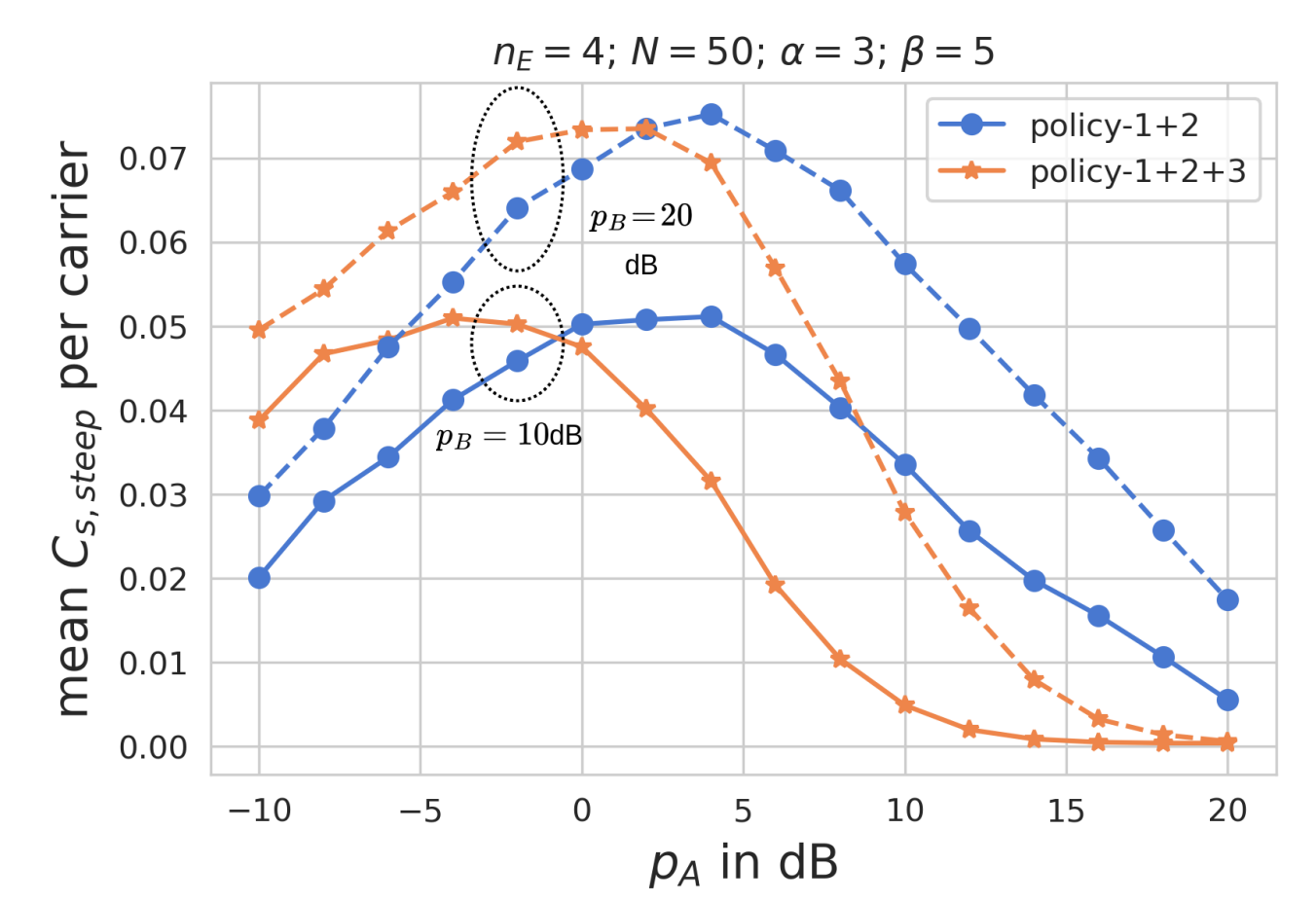
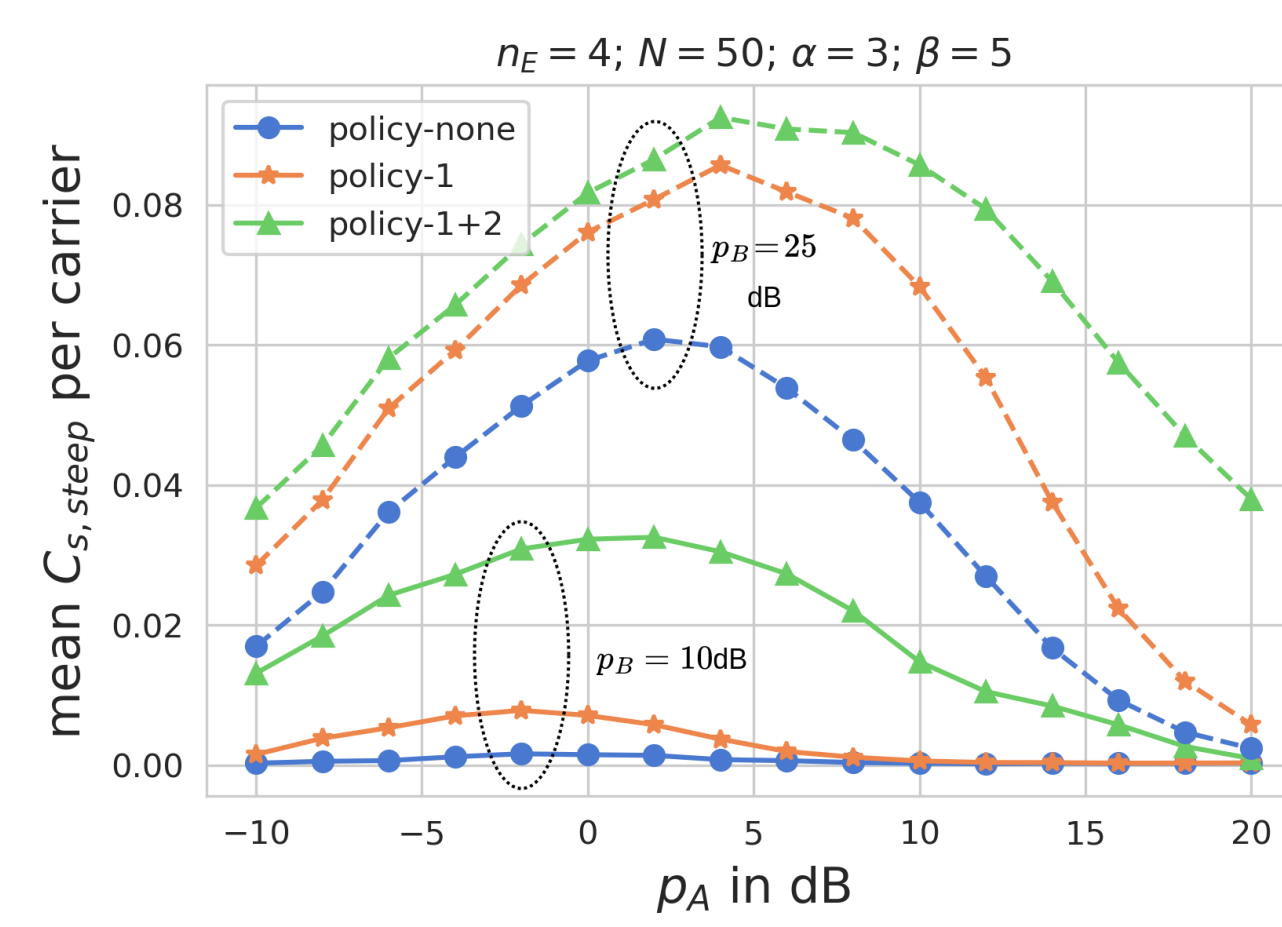
$$g'_n(p_{B,n}) = \mu; \forall n \text{ where } g'_n(p_{B,n}) > 0; \sum_{n=1}^N p_{B,n} = N p_B \quad (8)$$

- This can be efficiently solved using bisection search

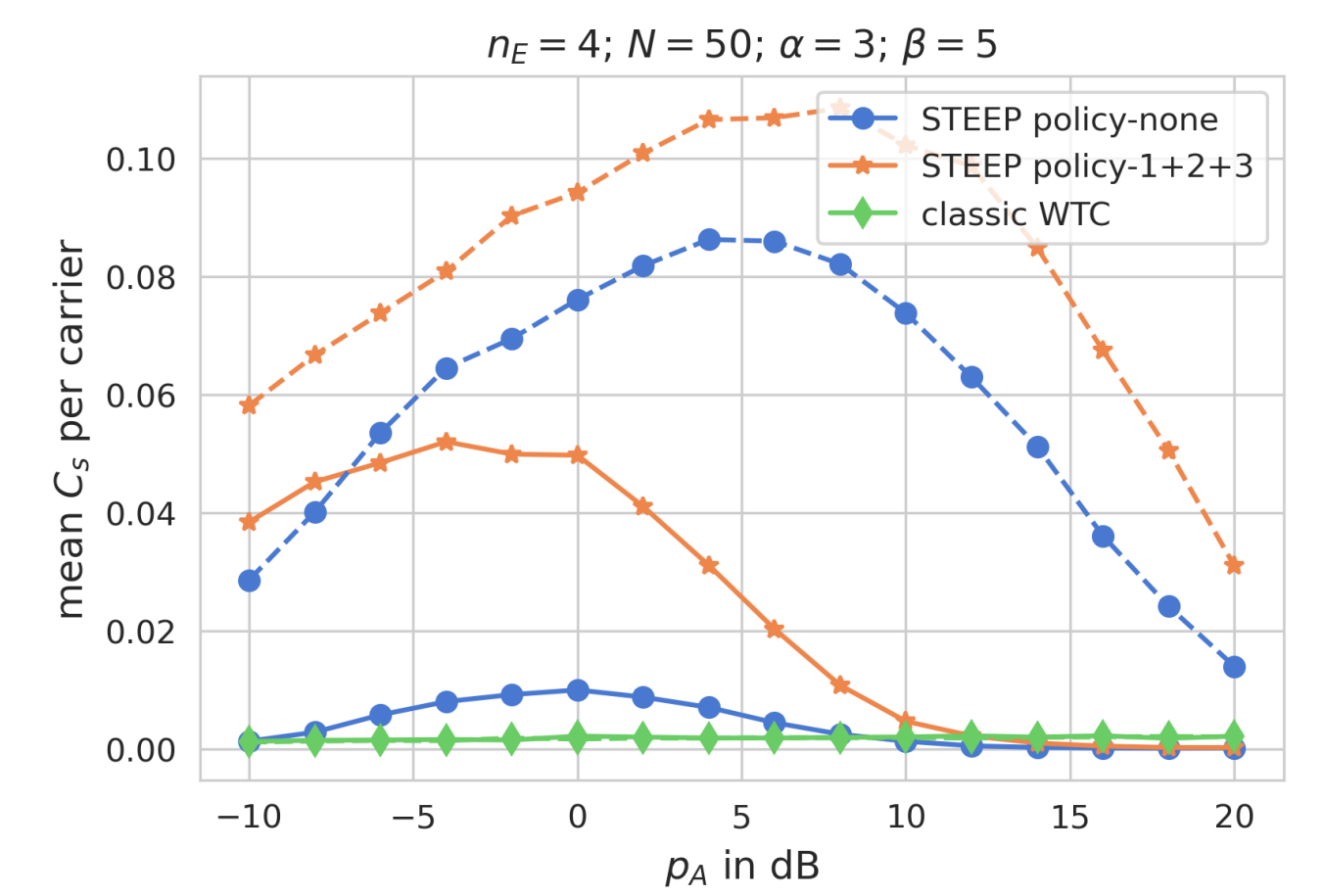
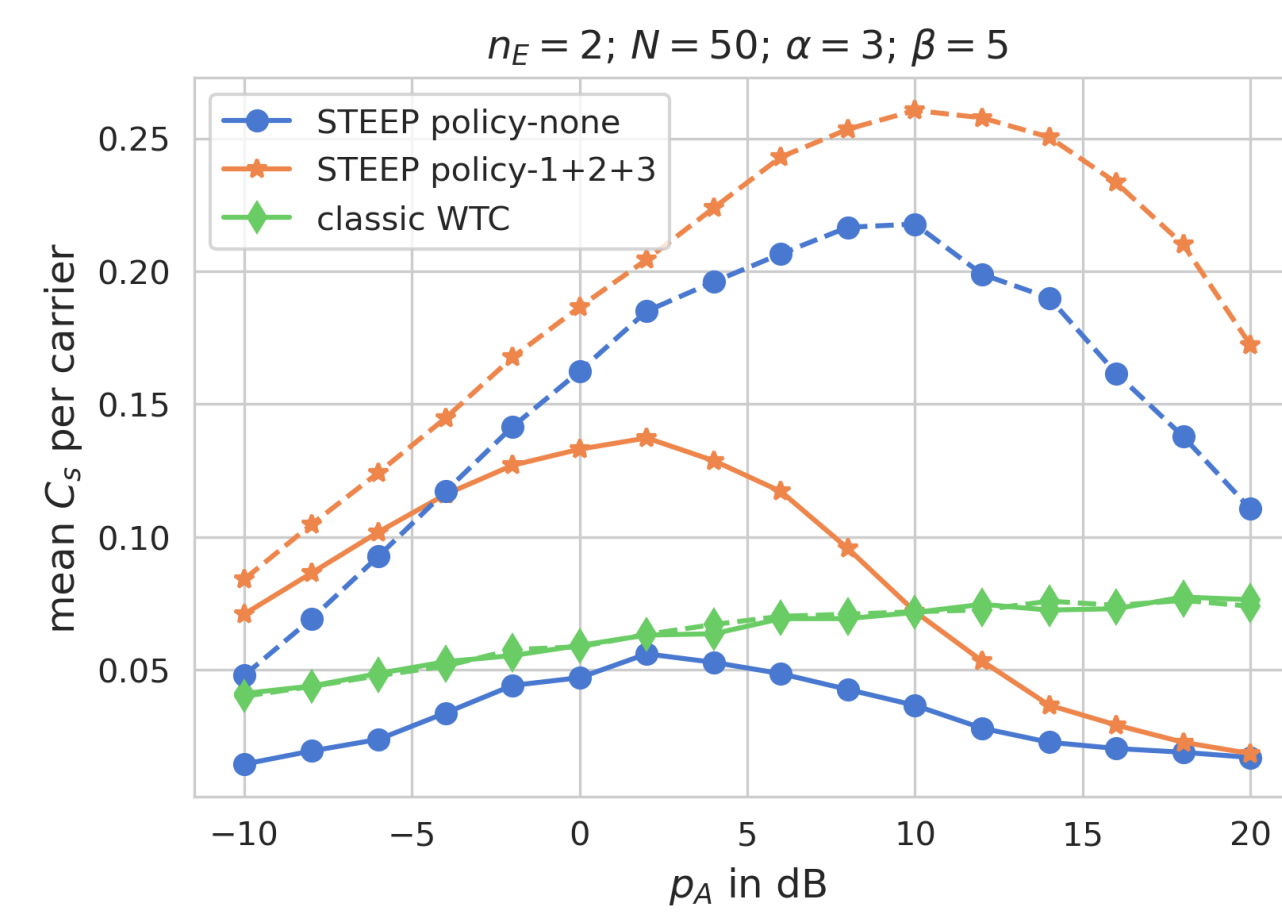
VII. Power Scheduling at Alice; policy-3

- Total power budget $N_C p_A$ for N_C carriers, power at carrier- n is $p_{A,n}$
- AASR given by (3) is not monotonically increasing w.r.t. $p_{A,n}$
- Without knowledge of Eve's CSI, no closed form
- After Bob allocate power to $N' \leq N$ echoing carriers, Alice equally distributes total power $N p_A$ only to the corresponding N' probing carriers, i.e., Alice also removes power from corresponding carriers where Bob removed power from and distribute total power equally to remaining carriers

VIII. Simulation results of policies



IX. Comparison with classic WTC channel



*The solid lines correspond to $p_B = 15$ dB and the dashed lines correspond to $p_B = 30$ dB

X. Discussion on Results

- Significant improvement is AASR by implementing policies-1,2,3
- Improved MC-STEPP performs significantly better than classic W.T.C. scheme in terms of secrecy
- AASR monotonically increases with echoing power but with probing power, AASR first increases then decreases.
- Implementing policy-3 does not improve AASR but saves power.

Reference

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- M. Hayashi and A. Vazquez-Castro, "Two-way physical layer security protocol for Gaussian channels," IEEE Transactions on Communications, vol. 68, no. 5, p. 3068–3078, May 2020.
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